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Dear Dr. Redondo and members of the Theoretical Physics Group,

My interest in theoretical physics has been driven by a desire to understand the fundamental structure of nature, which has naturally led me toward quantum field theory and particle physics. I am particularly fascinated by the role of symmetries in shaping physical laws, including how phenomena such as spontaneous symmetry breaking and anomalies play a central role in determining particle behaviour. I wish to deepen my understanding of these principles and ultimately piece the clues together to answer the question: what symmetries govern physics beyond the Standard Model?

My research experience spans fundamental particle physics, cosmology, and computational methods. Currently, I am working on my master's thesis, which focuses on gaining a deeper understanding of fundamental interactions and beyond-the-Standard-Model physics. In my research, I explore grand unified theories (GUTs). Rather than proposing a specific gauge group, I study a general landscape of high-energy GUTs and identify preferred models by minimizing their free energy. In parallel, I am involved in a project on infinite heat order, a framework related to high-temperature behaviour and symmetry non-restoration in UV-complete models. Through this work with my supervisor, Francesco Sannino, I have developed a strong foundation in quantum field theory.

Besides my thesis, my research experience includes a project on anomalies in quantum field theory. During my bachelor's degree, I worked on density profiles of dark matter, both theoretically and using machine learning techniques. This interest culminated in a bachelor's thesis in cosmology, where I examined whether inhomogeneities in the structure of the universe could explain its accelerated expansion. Across these projects, I have developed a strong foundation in theoretical physics and computational methods, which I am eager to apply to further study of fundamental questions.

For my PhD, I hope to contribute to research in theoretical particle physics. I yearn to use the framework of quantum field theory to study the interplay of flavour, Higgs dynamics, and beyond-Standard-Model physics, with the hope of uncovering the deeper principles governing fundamental interactions. Moreover, the opportunity to use machine learning and neural networks as tools for furthering our knowledge within theoretical particle physics excites me. Ultimately, I aim to contribute to clarifying the underlying principles that determine the structure of our universe.

I am strongly motivated to apply to the theoretical physics group at University of Zaragoza due to its broad department in theoretical physics, which provides an ideal setting for PhD training. I am particularly attracted to the research on particle phenomenology and beyond-the-Standard-Model physics. I would therefore be especially interested in learning from and contributing to research directions pursued by researchers such as S. Peñaranda and J. Alda.

Thank you for considering my application. I would be thrilled to contribute to the research activities of the theoretical physics group at University of Zaragoza.

Sincerely,

Sophie Lund Wagner